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Self-absorption correction method for one-point calibration laser-induced breakdown spectroscopy

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As an important variant of calibration-free laser-induced breakdown spectroscopy (CF-LIBS), one-point calibration LIBS (OPC-LIBS) corrects the Boltzmann plot of the unknown sample by using one known sample and obtains higher quantitative accuracy than CF-LIBS. However, the self-absorption effect restricts its accuracy. In this work, a new self-absorption correction (SAC) method for OPC-LIBS is proposed to solve this problem. This method uses an algorithm to correct the self-absorption and does not require the calculation of the self-absorption coefficient. To verify the effectiveness of this SAC method, Ti, V, and Al elements in two titanium alloys were determined by classical OPC-LIBS and OPC-LIBS with SAC. The average relative errors (AREs) of all elements in the two samples were decreased from 8.78% and 9.28% to 8.07% and 7.56%, respectively. The results demonstrated the effectiveness of this SAC method for OPC-LIBS. © 2022 Optica Publishing Group

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As the future superstar in analytical atomic spectrometry, laser-induced breakdown spectroscopy (LIBS) has many advantages including no or simple sample preparation and *in situ*, on-site, real-time, remote, and multi-elemental analysis. Meanwhile, LIBS can work in extreme environments. Thus, it has been applied in several fields such as metallurgical process monitoring [1], coal quality analysis [2,3], deep sea exploration [4], and Mars exploration [5].

Traditional LIBS quantitative analysis depends on calibration methods requiring several matrix-matched standard samples and building calibration curves, which is time-consuming and affected by the matrix effect [6]. Especially for the on-site and online analysis tasks, the fluctuations in the environment and changes in instrument performance dramatically affect the calibration process and restrict the LIBS quantitative accuracy [7]. In comparison, the calibration-free LIBS (CF-LIBS), proposed by Ciucci *et al.* in Ref. [8], determines the elemental concentrations using plasma parameters calculated by the Boltzmann

plot without standard samples. This method avoids building calibration curves, overcomes the matrix effect, and has great potential for *in situ*, on-site, and multi-elemental quantitative analysis of samples, for which standard samples are difficult to obtain.

However, the accuracy of CF-LIBS is often unsatisfactory. Therefore, many researchers have attempted to modify the CF-LIBS algorithm and several variants of CF-LIBS combined with the standard sample have been proposed [9]. As one of the most commonly used variants, one-point calibration LIBS (OPC-LIBS) was proposed by Cavalcanti *et al.* in Ref. [10]. Its main idea is to use one matrix-matched standard sample to correct the Boltzmann plot of the unknown sample. The authors tested OPC-LIBS on five bronze samples and obtained fairly good quantitative accuracy. Borduchi *et al.* used the Saha-Boltzmann plot to obtain a more accurate plasma temperature and higher accuracy of OPC-LIBS [11]. Fu *et al.* combined OPC-LIBS with the standard reference line method to overcome the lack of spectral lines of minor and trace elements [12]. Up to now, OPC-LIBS has many applications [13–15]. However, the self-absorption effect will decrease the accuracy of OPC-LIBS and no studies on this have been reported.

In our work, a new self-absorption correction (SAC) method for OPC-LIBS is proposed. This SAC method does not require the Stark coefficients of spectral lines to be known for the calculation of the self-absorption coefficient. We performed experiments on two titanium alloy samples. The results illustrated that this method can obtain higher quantitative accuracy than classical OPC-LIBS.

The main procedure of OPC-LIBS is described in detail by Cavalcanti *et al.* in Ref. [10]. Firstly, a matrix-matched standard sample is quantified using CF-LIBS based on the Boltzmann plot equation, expressed as

$$y_i = mx_i + q_s \quad (1)$$

$$\begin{cases} x_i = E_i \\ y_i = \ln \frac{I_{ij}}{A_{ij}g_i} \end{cases} \quad (2)$$

$$\begin{cases} m = -\frac{1}{K_B T} \\ q_s = \ln \frac{FC}{U_s(T)} \end{cases}, \quad (3)$$

where I_{ij} is the line intensity; A_{ij} is the transition probability between the two levels; E_i and g_i are the energy and degeneration of the upper level, respectively; T is the plasma temperature; $U_s(T)$ is the partition function at plasma temperature T ; C is the concentration of the species s ; F is an experimental parameter; and K_B is the Boltzmann constant.

According to Eqs. (1)–(3), the plasma temperature T can be estimated by linear regression of the points, representing spectral lines of the same species in the $x_i y_i$ plane. The experimental parameter F and the concentration C of each element in the standard sample can be calculated through the T and normalized relationship of concentrations (the sum of the concentrations of all species is 100%). Then, we will find that the calculated concentrations C will be different from the certified ones C_{cert} . This means that the points do not locate where they should be in the Boltzmann plot due to the offsets of the y_i coordinate values. The offset Δy_i and a correction parameter P_i can be calculated by

$$y'_i = -\frac{E_i}{K_B T} + q_s + \ln \left(\frac{C_{cert}}{C} \right) \quad (4)$$

$$\Delta y_i = y'_i - y_i \quad (5)$$

$$P_i = A_{ij} g_i \cdot e^{-\Delta y_i}. \quad (6)$$

Before the unknown sample is quantified via the CF-LIBS procedure, we can modify $y_i = \ln[I_{ij}/(A_{ij} g_i)]$ in Eq. (2) to $y_i = \ln(I_{ij}/P_i)$ to eliminate the offsets in advance. According to the results in the work of Cavalcanti *et al.*, the uncertainty of the transition probability and the differences in spectral acquisition efficiency can be eliminated through the correction parameter P_i [10].

It should be noted that the correction parameter P_i can correct self-absorption to some extent. During the execution of OPC-LIBS, the calculated elemental concentrations of the standard sample by CF-LIBS will be affected by self-absorption. The self-absorption will make the measured values of spectral intensities lower than theoretical values, so the y_i coordinate values of the points in the Boltzmann plot will be lower than the theoretical values. The offsets, corrected by the parameter P_i , actually contains three sources: the uncertainty of the transition probability; the differences in spectral acquisition efficiency; and the possible self-absorption. In other words, the correction parameter P_i contains the self-absorption coefficient SA if self-absorption exists. The first two are independent of the elemental concentration, but the last one is not. The SA value varies with the linewidth, which varies with the elemental concentration under the self-absorption effect. Therefore, in the OPC-LIBS algorithm, the influence of the first two can be reasonably eliminated, whereas self-absorption cannot. Self-absorption needs to be corrected unless the self-absorption of the target lines can be ignored, or the target lines have the same extent of self-absorption.

According to the self-absorption theory, the self-absorption coefficient SA can be expressed as [16]

$$(SA)^{0.5} = \frac{\Delta \lambda_0}{\Delta \lambda} \approx \frac{I \Delta \lambda}{I_0 \Delta \lambda_0}, \quad (7)$$

where $\Delta \lambda$ and I are the measured values of the linewidth and intensity, respectively, and $\Delta \lambda_0$ and I_0 are the theoretical values

of the linewidth and intensity unaffected by the self-absorption, respectively.

Rearranging the terms, Eq. (7) can be expressed as

$$I_0 = I \left(\frac{\Delta \lambda}{\Delta \lambda_0} \right)^2. \quad (8)$$

$\Delta \lambda_0$ can be calculated by the electron number density N_e , described as

$$\Delta \lambda_0 = 2\omega \frac{N_e}{N_e^{ref}}, \quad (9)$$

where ω is the electron impact parameter and is constant under certain conditions. N_e^{ref} is the reference electron density, generally 10^{16} cm^{-3} . N_e can be estimated by [17]

$$N_e = 8.02 \times 10^{12} \left(\frac{\Delta \lambda_{H\alpha}}{\alpha_{1/2}} \right)^{3/2}, \quad (10)$$

where $\Delta \lambda_{H\alpha}$ is the Lorentzian full width at half maximum (FWHM) of the $H\alpha$ line, and $\alpha_{1/2}$ is the half width of the reduced Stark profile, which is described by Griem in Ref. [18]. In LIBS spectra, the $H\alpha$ line comes from water in the air, and its self-absorption does not occur due to the low concentration.

Combining Eqs. (8)–(10), I_0 can be rewritten as

$$I_0 = I(\Delta \lambda)^2 (\Delta \lambda_{H\alpha})^{-3} (\alpha_{1/2})^3 \left(\frac{N_e^{ref}}{1.604 \times 10^{13} \omega} \right)^2. \quad (11)$$

Then Eq. (11) can be further rewritten as

$$W = (\Delta \lambda)^2 (\Delta \lambda_{H\alpha})^{-3} \quad (12)$$

$$Q = (\alpha_{1/2})^3 \left(\frac{N_e^{ref}}{1.604 \times 10^{13} \omega} \right)^2 \quad (13)$$

$$I_0 = I \cdot W \cdot Q, \quad (14)$$

where Q is a constant for the same spectral lines, W contains two variables—the measured width of the target line and $H\alpha$ line—and I is the measured intensity of the target line.

According to the previous analysis, parameter P_i can be expressed as $P_i = R_i/(W_i \cdot Q_i)$ for the optically thick lines, where R_i is a correction factor for the uncertainty of the transition probability and the differences in spectral acquisition efficiency, and $(W_i \cdot Q_i)$ is a correction factor for SAC. The non-constant term W_i can be calculated for the standard and the unknown samples, respectively. Thus, we can define another correction parameter P_i^{SAC} for the optically thick lines, expressed as

$$P_i^{SAC} = \frac{R_i}{Q_i} = P_i \cdot W_i. \quad (15)$$

When conducting the CF-LIBS part in OPC-LIBS, we need to calculate the W_i values of the optically thick lines of the standard sample. Then P_i and P_i^{SAC} can be calculated by Eqs. (6) and (15) for the optically thin and optically thick lines, respectively. When quantifying the unknown sample by the modified Boltzmann plot equation, we also need to calculate the W_i values of the optically thick lines and use $y_i = \ln(I_{ij}/P_i)$ and $y_i = \ln(W_i \cdot I_{ij}/P_i^{SAC})$ for the optically thin and optically thick lines, respectively. It should be noted that the SAC algorithm can also be used for all spectral lines. The W_i values of optically thin lines from the standard and test samples will be the same and only optically thick lines will be corrected. But the width calculation by fitting line is time-consuming and we suggested performing the SAC algorithm on the elements with few lines.

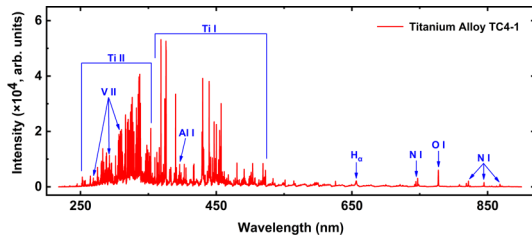


Fig. 1. Ten averaged LIBS spectra of the titanium alloy sample TC4-1.

Table 1. Selected Analytical Spectral Lines Used for CF-LIBS and OPC-LIBS

Species	Wavelength (nm)
Ti I	363.55, 365.35, 395.63, 395.82, 398.18, 399.86
Ti II	253.13, 253.46, 253.59, 255.60, 257.10, 263.54, 263.86, 264.20, 264.59, 295.88, 309.72, 314.80, 316.85, 319.09, 320.25, 328.77, 345.25, 345.64, 346.15, 347.72, 351.08, 352.03, 353.54, 359.60, 374.16, 390.05
V II	296.84, 300.12, 311.07, 313.97, 326.77, 354.52, 355.68
Al I	394.40, 396.15

The LIBS system used in this work is the same as that previously reported [19]. A Q-switched Nd: YAG laser (Quantel, Q-smart 850) was used to ablate the sample and generate the plasma, operating at a wavelength of 532 nm with an energy of 100 mJ and a repetition rate of 5 Hz. An echelle spectrometer (Andor tech., Mechelle 5000) coupled with an intensified CCD camera (Andor tech., iStar DH334T) was used for the acquisition of LIBS spectra, employing a gate delay of 2 μ s and a gate width of 500 ns. Ten spectra were acquired and averaged per sample. Each spectrum was accumulated by ten laser pulse signals to increase the stability of spectral intensity and signal-to-noise ratio.

To verify the SAC algorithm described above, we used three titanium alloy TC4 samples (GBW02503/05/07, No.12 Institute, China Shipbuilding Industry Group Co., Ltd.) along with certified reference materials. We considered TC4-1 as the standard sample, while TC4-2 and TC4-3 were considered as unknown and tested by CF-LIBS, OPC-LIBS, and OPC-LIBS with SAC. Figure 1 shows the typical LIBS spectrum of TC4-1. The selected analytical spectral lines are listed in Table 1. The transition probabilities and energy and degeneracy of the upper level of lines can be obtained from the National Institute of Standards and Technology atomic spectra database [20].

TC4-2 was selected as an example. Figure 2(a) shows the Boltzmann plot of TC4-2, determined by CF-LIBS. We used the ionic lines of Ti to fit the straight line in the Boltzmann plot and calculate the plasma temperature. The slope of the fitted line of Ti II was used as the line slope for other species so that each species could have a straight line in the Boltzmann plot and the CF-LIBS process could be completed.

As shown in Fig. 2(a), it is obvious that the points are scattered on both sides of the fitted line, especially for Ti, due to uncertainty of the transition probability, the differences in spectral acquisition efficiency, and possible self-absorption. The concentrations of all three elements in titanium alloys are so high that the self-absorption cannot be ignored. For Ti and V, we chose as few resonance and persistence lines as possible. However, non-resonance and non-persistence lines tend

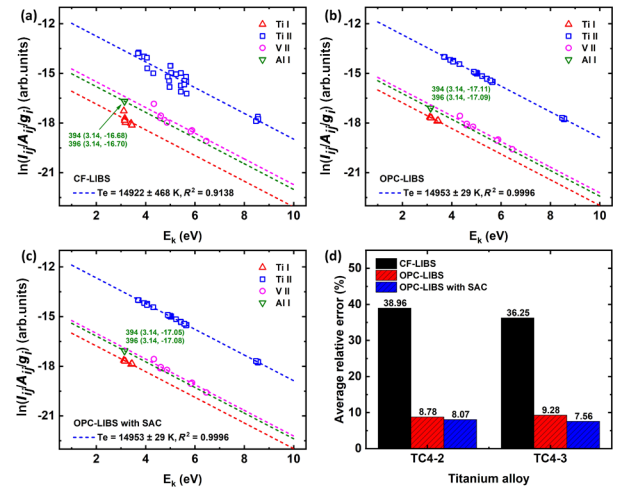


Fig. 2. (a) Boltzmann plot of CF-LIBS; (b) modified Boltzmann plot of OPC-LIBS; (c) modified Boltzmann plot of OPC-LIBS with SAC; (d) quantitative AREs of titanium alloy samples TC4-2 and TC4-3.

Table 2. Quantitative Results of Titanium Alloy Samples

TC4	Methods	Elemental Concentration (wt%)					
		Ti	RE (%)	V	RE (%)	Al	RE (%)
2#	Reference	91.17	/	3.43	/	5.40	/
	CF-LIBS	89.97	1.32	6.32	84.26	3.71	31.30
	OPC-LIBS	91.59	0.46	3.81	11.08	4.60	14.81
	OPC-LIBS with SAC	91.48	0.34	3.81	11.08	4.71	12.78
3#	Reference	89.34	/	3.86	/	6.80	/
	CF-LIBS	89.54	0.22	6.46	67.36	4.00	41.18
	OPC-LIBS	90.97	1.82	3.91	1.30	5.12	24.71
	OPC-LIBS with SAC	90.67	1.49	3.90	1.04	5.43	20.15
1#	Standard	90.37	/	5.70	/	3.93	/

to have a high uncertainty of the transition probability. For Al, only two resonance lines were detected—Al I 394.40 and 396.15 nm—predicting undesired quantitative results.

Figure 2(b) shows the modified Boltzmann plot, determined by OPC-LIBS. Most of the points are distributed on the fitted lines and the values of the determination coefficient R^2 of the Ti fitted lines are both larger than 0.999. The calculated values of the plasma temperature by OPC-LIBS are similar to those by CF-LIBS. This means that the OPC method cannot correct the deviation of the temperature estimation and mainly improves the quantitative accuracy by increasing R^2 and correcting the intercepts of the fitted lines. Besides, some points are not fully corrected in the modified Boltzmann plot, but this does not affect the accurate fitting of the line and subsequent quantification.

For OPC-LIBS with SAC, we performed the SAC algorithm on Al. As shown in Fig. 2(c), the y_i coordinate values of Al lines were increased compared with those in Fig. 2(b), which meant that self-absorption was well corrected. The detailed quantitative results are listed in Table 2. For OPC-LIBS with SAC, the relative errors of Al were reduced from 14.81% and 24.71%

to 12.78% and 20.15% for TC4-2 and TC4-3, respectively. As shown in Fig. 2(d), the average relative errors (AREs) of OPC-LIBS after SAC were reduced from 8.78% and 9.28% to 8.07% and 7.56% for TC4-2 and TC4-3, respectively. These results showed that the proposed SAC method was effective for the accuracy improvement of OPC-LIBS.

To sum up, a new SAC method for OPC-LIBS is proposed. This method uses an algorithm to correct self-absorption and does not require the calculation of the self-absorption coefficient. The SAC algorithm was described in detail and titanium alloy samples were used to verify its effectiveness. The drop in quantitative relative errors is obvious. The results illustrated that this SAC method can further improve the quantitative accuracy of OPC-LIBS. These further promote the practical application of OPC-LIBS and CF-LIBS, which is significant for online and on-site multi-elemental quantitative analysis.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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